

Inter-frame forgery detection and localisation in videos using earth mover's distance metric

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Abstract: Video forensics is one of the hot topics in multimedia forensics. Nowadays, spreading fake videos across the internet is becoming a profession mainly in politics and entertainment. In this study, a novel two-stage inter-frame video forgery detection technique is proposed. The first stage analyses spatio-temporal feature flow consistency to detect suspicious tamper points. Identifying the type of forgery and validating the recovered video are done in the second stage. Earth mover's distance is used as a similarity metric in both stages. The authors concentrate on a robust inter-frame forgery detection approach which can be applied for any challenging video. Compression at a higher rate, noise addition, and filtering are the anti-forensic tricks used by forgers to fool forensic techniques. However, most of the literature in video forgery detection has handled these issues as post-processing attacks and reported lesser accuracies for it. Hence the authors propose a robust and efficient forgery detection technique capable of identifying all kinds of inter-frame forgeries in videos. Experimental evaluation of the public video data set shows that the proposed approach outperforms existing approaches with an improved rate of robustness.

1 Introduction

Recent advancements in video processing technology led to the immense spread of videos across the internet. Lots of people adore posting videos on social media applications like YouTube, WhatsApp, Facebook etc. However, the candidness of these posted videos is highly uncertain. Such deceptive videos populated in social media may detriment the harmony of society. Also, malicious tampering of videos creates legal problems as videos are used as evidence in the court of law. Consequently, video forensics has emerged to ascertain the forgery in a video file.

Most of the digital forensic approaches mainly concentrate on images. However, recent researchers have focused on video forensics due to the massive increase in video tampering. Generally, video forgery can be classified into intra-frame forgery and inter-frame forgery. Intra-frame forgery involves region-based forgeries that occur within a frame in the spatial domain. Popular intra-frame video forgeries are object clipping from a different video [1], object removal [2], object addition [3], copy-move video forgery [4], and moving foreground removal [5]. Matured image forensic techniques such as contrast manipulation detection [6], geometrical transformation detection [7, 8], and copy-paste or copy-move forgery detection [9–11] can be adopted for intra-frame forgery detection as a video is a collection of images.

Forgeries like frame insertion, frame deletion, and frame duplication come under the second category called inter-frame forgery which occurs in the time domain. Frame insertion augments a sequence of frames at some location in a video file. The augmented frames may be taken either from the same video (which is known as frame duplication) or foreign video. Inter-frame attacks aim to extend or hide an event in a video. This kind of forensic technique employs temporal features across the frames rather than image-level features within a frame for tamper detection purpose. Inter-frame forgeries are simple and preferable for forgers as it involves only copy-paste/delete a set of frames, whereas intra-frame forgery involves a lot of work, i.e. to edit one second of a video, forgers need to touch ~25 frames.

The organisation of our paper is as follows: the related work for inter-frame video forgery detection is outlined in Section 2. In Section 3, we describe earth movers distance (EMD) which is the preliminary of the proposed work. We elucidate the proposed methodology in Section 4 and the experimental results are presented in Section 5. Finally, the work is concluded in Section 6.

2 Related work

In this section, we explore some of the related works for inter-frame forgery detection in videos. Most of the inter-frame video forensics come under two aspects of detection strategies – one is based on the consistency of a video feature; the other one is related to artefacts introduced by video coding standards.

2.1 Video feature based methods

The basic idea of these methods is that a distinct video feature is continuous, whereas the continuity would be broken for tampered videos. Yang *et al.* [12] proposed a method for frame duplication detection using singular value decomposition (SVD) features. The SVD features of each frame were extracted and a similarity analysis was performed with all other frames using Euclidean distance. Similarly, Ulutas *et al.* [13] extracted binary features from video frames and performed a similarity measure among the features for frame duplication detection. Since these methods considered the features of all the frames for similarity comparison, they are time-consuming. Fadl *et al.* [14] proposed an efficient method for frame duplication by calculating the entropy of discrete cosine transform coefficients of residual frames and ignoring other frames. Although this is time-efficient, these kinds of similarity analysis algorithms can be applied only for duplication attack. Zhang *et al.* [15] constructed motion vector pyramid (MVP) for adjacent frames to locate the discontinuity point. MVP is good at locating the discontinuity point but the correlation between MVPs of similar frames is not so strong, which resulted in poor identification accuracy for frame duplication. Jia *et al.* [16] handled copy-move video forgery in his work. The authors extracted optical flow information as a temporal feature and tested its consistency for tampering detection. The optical flow correlation is then used for finding the duplicated frame pairs. This work reported lesser accuracy for compressed videos. Moreover, this method fails for a carefully prepared copy-move forgery with largely static scenes. Li *et al.* [17] extracted the two-dimensional (2D) phase congruency of each frame and calculated the correlation among the adjacent frames. Also, the authors used the *k*-means clustering algorithm for categorising normal and abnormal points. Since this method does not perform similarity analysis of tampered frames with remaining frames, it fails to report whether the tampered frames were taken from the same video or a different

source video. Fadl *et al.* [18] extracted energy residue of each frame from a video and determined abnormal points in the energy flow. Thompson tau test [19] is used for anomaly detection in their work. Although the authors reported good accuracy and robustness for compression, the method is not tested for small lengths forgery, i.e. inserting/deleting a few frames in a video. The general drawback of feature-based methods is that most of the works are not generic to all inter-frame forgery attacks and are not robust to compression attack.

2.2 Video codec based methods

Video codec based forensic techniques focus on identifying forgeries embedded in compressed videos under the assumption that forgers usually edit a compressed video by decompressing it followed by recompression. These techniques reveal the traces left behind during the recompression of an edited video. These techniques assume that multiple compressions in a video are a potential clue for edited videos. Stamm *et al.* [20] developed a mathematical model to detect frame insertion and frame deletion by using video's P -frame prediction error sequence which is caused by the amendment in the group of pictures (GOPs) due to double compression. He *et al.* [21] obtained double compression fingerprints using local motion vector field of moving objects in a video. According to the strength of the estimated motion fields, the reformed prediction residual sequence is calculated. Bestagini *et al.* [22] exploited codec adopted and size of GoPs used in the coding steps for double compression. Aghamaleki and Behrad [23] studied the traces of quantisation error on residual errors of the P -frames to detect and localise forged frames. Feng *et al.* [24] studied the characteristics of interfering frames and identified frame deletion points based on frame motion residuals. The main drawback of these works is that they depend on the codec used for video compression. Moreover, videos compressed using lossless compression are challenging for these methods.

To sum up, most of the video feature based methods cannot be applied to all types of inter-frame forgery like frame insertion, frame deletion, and frame duplication. Moreover, they are not robust to attacks such as compression, noise addition, and filtering following the tampering process of videos. Since most of the real-time videos are compressed heavily for storage purposes, a full-bodied video forensic is a necessity of the day. On the other hand, video codec based methods can be applied for a video compressed using a particular coding standard. Also, these methods become problematic for lossless compression and very high-quality videos. Hence our focus is on detecting all the three inter-frame forgeries in both compressed and uncompressed videos without compromising for accuracy.

3 Preliminaries

In this section, we brief about EMD [25], which is used as a correlation metric in our work. EMD is an optimal metric between two distributions which matches the perceptual similarity. It estimates the minimal cost to be invested to convert one distribution to another. The distributions represent different features of an image and can be of fixed or variable size. EMD is mainly used in content-based image retrieval [26] and is more robust and efficient than many other similarity metrics like Euclidean distance (as used in [12, 13] for similarity check) and mean correlation (as given in [14, 16] for comparison between frames).

EMD can be intuitively thought of the amount of work needed to fill holes in the earth (one distribution) to match a mass of earth properly spread in space (another distribution). In other words, the minimum work needed to transform one distribution into the other. The computations are based on a well-known transportation problem.

EMD problem can be mathematically formulated as a linear programming problem. Let P and Q represent two signatures (set of feature clusters) with m and n number of cluster features, respectively.

$$P = \{(p_1, w_{p1}), \dots, (p_m, w_{pm})\} \quad (1)$$

$$Q = \{(q_1, w_{q1}), \dots, (q_n, w_{qn})\} \quad (2)$$

where p_i and q_j are the cluster representatives (mean or mode) and w_{pi} and w_{qi} are the cluster weights of P and Q . Here cluster weights denote the fraction of pixels belonging to a cluster. The initial ground distance between two representative features aggregated in a cluster is defined by a matrix, $D = [d_{ij}]$, where d_{ij} denotes the ground distance between clusters p_i and q_j . The ground distance can be any distance chosen depending on the application, such as distance between individual colours in case of colour distribution and L_1 in case of distance between texture features etc. Let $F = [f_{ij}]$ be the flow matrix, where f_{ij} denotes the optimal flow between p_i and q_j with minimum cost. The problem can be devised mathematically as formulated in (3)–(7):

$$\text{WORK}(P, Q, F) = \sum_{i=1}^m \sum_{j=1}^n d_{ij} f_{ij} \quad (3)$$

subject to the following constraints:

$$f_{ij} \geq 0 \quad 1 \leq i \leq m, 1 \leq j \leq n \quad (4)$$

$$\sum_{j=1}^n f_{ij} \leq w_{pi}, \quad 1 \leq i \leq m \quad (5)$$

$$\sum_{i=1}^m f_{ij} \leq w_{qj}, \quad 1 \leq j \leq n \quad (6)$$

$$\sum_{i=1}^m \sum_{j=1}^n f_{ij} = \min \left(\sum_{i=1}^m w_{pi}, \sum_{j=1}^n w_{qj} \right) \quad (7)$$

Constraint (4) restricts the flow movement from P to Q and not vice versa. Constraint (5) limits the flow amount that can be sent by clusters in P by their weights. Similarly, constraint (6) limits the flow amount received by clusters in Q to be no more than their weights. Finally, constraint (7) forces the maximum possible flow amount which is called total flow. Once solving the linear programming problem, the optimal flow, F would be found. The EMD defined in (8) can now be derived from (3) as the work normalised by the total flow

$$\text{EMD}(P, Q) = \frac{\sum_{i=1}^m \sum_{j=1}^n d_{ij} f_{ij}}{\sum_{i=1}^m \sum_{j=1}^n f_{ij}} \quad (8)$$

A normalisation factor is needed when two signatures have different total weights in their respective regions. The total weight of the smaller signature is chosen as a normalisation factor to avoid favouring a smaller signature.

The advantages of EMD over other distribution distances are as follows: (i) feature vectors described as signatures are more compact and flexible. EMD can also handle variable-sized signatures, (ii) most of the other similarity measures incurs quantisation problem which is avoided in EMD by using the minimised transforming cost property (iii) EMD supports partial matches to deal with occlusion, clutter, and mediocre quality images (iv) EMD is an optimal measure which endows pixel space with a true metric structure.

4 Proposed methodology

The proposed work is based on the regularity and continuity of original video features. If the video has tampered through insertion, deletion, or duplication of selected frames the regularity and continuity will be broken. In this work, we initially extract the EMD between adjacent frames in a video. If the video is original,

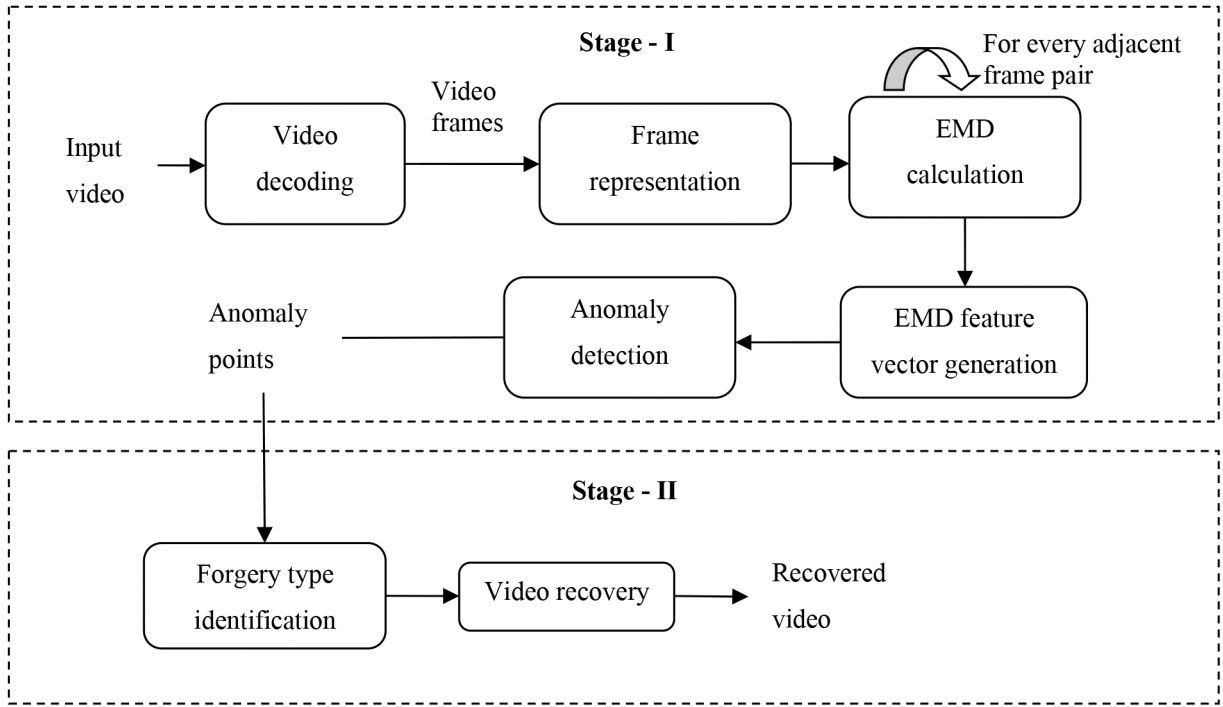


Fig. 1 Overview of the proposed system

the EMD flow is consistent, otherwise, the flow shows anomalies. The flow of methodology is depicted in Fig. 1.

In the first stage, each frame is represented as spatio-temporal feature representation. The EMD metric between features of every adjacent frame is estimated and converted into a vector. The sudden changes in this EMD vector are identified as anomaly points. Stage II analyses the anomaly points and discovers the type of forgery. Also an entire list of tampered frames are located and the video is recovered in this stage.

4.1 Video decoding

Video is a collection of frames. These frames have to be extracted before processing. Hence the first step is to decode the input video file and extricate it into frames.

4.2 Frame representation

Each frame is then represented as a feature vector. We employ joint feature distribution consisting of pixel brightness (in the spatial domain) and moving pixel position (in the temporal domain). Pixel brightness flow in a video across the frames is steady and is a good discriminator of two non-adjacent frames. The brightness content of each frame is described by a 1D distribution of image intensities in the case of grey-scale video, and as a 3D colour space distribution for colour videos. The frames of the input video in RGB format are smoothed first to reduce colour quantisation and dithering artefacts. RGB frames are then converted to CIE-Lab colour space representation. CIE-Lab colour space is adopted for its power of expressiveness and perceptual uniformity.

We also consider the position of moving pixels into account, as sometimes the motion distribution for two frames might be similar, but the position of moving objects in the frames is different. The direction and speed of moving objects in a video are estimated using the optical flow method defined by Lucas and Kanade in [27], and the corresponding moving pixel positions are estimated. The pixel coordinates are normalised to be in the range of 0 to 100. Hence the joint feature representation is a 5D space as defined as

$$F = \{(L, a, b), (x, y)\} \quad (9)$$

where F denotes the feature distribution of a frame, (L, a, b) symbolises a set of dominant colour values in CIE-Lab colour space, and (x, y) represents coordinates of each moving pixel. The

joint feature distribution in (9) is then coalesced into clusters of similar features to represent frames as signatures as defined below

$$S_f = \{s_i = (m_i, w_i)\} \quad (10)$$

where S_f is the signature of frame f , s_i is the signature of the i th cluster in frame f , m_i represents the mean of the i th cluster, and w_i represents the fraction of pixels in the frame belonging to the i th cluster. We use sensitivity-based clustering [28] to cluster the distribution and we remove very small clusters with insignificant weights ($<0.1\%$). The number of clusters in a signature is not fixed because the amount of significant information in a video frame depends on the content of the video. For instance, when considering motion information, a football video has a large magnitude of movement from one frame to another frame, and hence a finely quantised signature is needed. On the other hand, a news reading video has less motion change and so a coarsely quantised signature would be sufficient. In brief, a balance between expressiveness and efficiency cannot be achieved in fixed-sized signatures.

4.3 EMD calculation

Given two signatures, a quantitative dissimilarity measure approximating perceptual dissimilarity is defined using EMD. Defining a distance metric between two frame signatures requires the ground distance between basic features that are aggregated into clusters. We use L^2 norm distance between basic signatures as the underlying ground distance. In doing so, the ground distance matrix is denoted as $D = [d_{ij}]$, where d_{ij} is formulated in (11) which calculates L^2 norm distance between two signatures s_i and s_j .

$$\sqrt{(\Delta L)^2 + (\Delta a)^2 + (\Delta b)^2 + \lambda((\Delta x)^2 + (\Delta y)^2)} \quad (11)$$

The parameter λ in (11) specifies the weightage given to moving pixel locations. When we set $\lambda = 0$, motion information will be ignored. We suggest this setting for stationary videos with no moving objects in it. The colour information alone will be sufficient enough to locate tampered frames in stationary videos. In general, λ is set to 1 to give equal weightage to spatial and temporal information. The EMD score value between two signatures is computed to compare them as illustrated in Section 3. Two signatures have an EMD score equal to zero if they are the

```

 $L = detectAnomaly(V, T_{peak})$ 
//Input : Input Video,  $V$  and threshold,  $T_{peak}$ 
//Output : List of anomaly frame numbers,  $L$ 
1:  $v = readVideo(V)$ 
2:  $nof = v.numberofFrames$ 
3: for  $i=1:nof-1$  do
4:    $E(i) = emd(v(i), v(i+1))$ 
5: end for
6: for  $i=1:nof-1$  do
7:    $\alpha_i = \frac{E(i)}{\frac{1}{4}(E(i-2) + E(i-1) + E(i+1) + E(i+2))}$ 
8:   if  $\alpha_i > T_{peak}$  then
9:     add  $i$  to  $L$ 
10:  end if
11: end for

```

Fig. 2 Algorithm 1

representation of the same frames. In general, adjacent frames in a video have EMD score nearer to zero and are minimal for the same shot.

4.4 EMD feature vector generation

The EMD vector, E of size $N-1$, for a video sequence with N frames is formed by computing the EMD score value of every two adjacent frames in the video as follows:

$$E = [EMD(S_1, S_2), EMD(S_2, S_3), \dots, EMD(S_i, S_{i+1}), \dots, EMD(S_{N-1}, S_N)] \quad (12)$$

where $EMD(S_i, S_{i+1})$ denotes EMD value between signatures of i th and $(i+1)$ th frames.

4.5 Anomaly detection

EMD feature vector E should be continuous for an original video. To determine outliers in the vector, we compute the rate of change of every element in E . The rate of change α_i in (13) defines the variation extent of i th EMD score to determine whether it is a sudden change or not. A larger α_i indicates that the EMD score of i th frame varies a lot from its nearest neighbours. We have considered four neighbours of i th frame – $(i-2)$, $(i-1)$, $(i+1)$, $(i+2)$. We use a peak threshold T_{peak} to locate peaks (anomalies) in the plot of E . If α_i is larger than T_{peak} , it means that i th frame is showing a sudden spike from its neighbours

$$\alpha_i = \frac{E(i)}{(1/4)(E(i-2) + E(i-1) + E(i+1) + E(i+2))} \quad (13)$$

The anomaly detection process is summarised in Algorithm 1. Line 1 creates an object v for reading input video V . Line 2 extracts the number of frames information from v and stores it in a variable nof . Lines 3–5 calculate EMD vector E for every adjacent frames $v(i)$ and $v(i+1)$. The function $emd()$ accepts two images and finds a score for the signature representation of those two images based on their similarity. If two images are the same, then $emd()$ returns zero. Line 7 estimates the rate of change α_i for every element of E . Lines 8–10 checks α_i with threshold T_{peak} and adds to anomaly frame list L if $E(i)$ is a spike (see Fig. 2).

4.6 Forgery type identification

To identify the type of inter-frame forgery from the EMD plot, we need to analyse the plot. Three possible tampering situations may occur – frame insertion, frame deletion, and frame duplication. If two distinct peaks are produced in the EMD plot, then the video is attacked by either insertion or duplication forgery. This is because insertion and duplication forgery have a start and end tampering point, whereas deletion forgery comprises a single deletion point. However, sometimes two striking peaks in the plot may be due to repeated deletion of frames at various timelines of a video. To distinguish these two situations, we find the EMD correlation score between the previous frame of the first peak and the next frame of the second peak. Assume that frames from i th location to j th location are tampered frames due to frame insertion/duplication into a video. Then $(i-1)$ th and $(j+1)$ th frames are originally adjacent to each other. Hence frame pair $(i-1, j+1)$ has a high EMD correlation due to adjacency and also contributes to the continuous EMD flow after eliminating EMD values from i to j (since they are the EMD values of tampered sequence). This adjacency and continuity are not noticeable in case of deletion forgery. To differentiate insertion and duplication forgery, we use a duplicate frame pair matching algorithm (Algorithm 2). Forgery type identification steps are detailed in Algorithm 3. Algorithm 1 lists the anomaly frames producing sudden spikes in the EMD plot as the start or endpoint of a suspicious frame sequence. Algorithm 3 identifies the type of forgery as frame insertion or frame deletion or frame duplication on analysing the anomaly frames. Line 1 of Algorithm 3 iterates for all frames in the anomaly frame list L . Line 2 finds the EMD correlation between the previous frame of the i th anomaly frame and the next frame of the $(i+1)$ th anomaly frame. If the correlation score is less than the threshold, T_{corr} , then both the frames i and $i+1$ are most probably adjacent frames. This case occurs for frame insertion and frame duplication. The frames from i to $i+1$ are reported as tampered frames and are stored in L' . To differentiate frame insertion and frame duplication, Algorithm 2 is used. If Algorithm 2 returns duplicate frame pairs, then duplication forgery has taken place, otherwise, insertion forgery has happened. Line numbers 5–9 perform this checking and line 10 extracts the first column of D , which contains duplicated frame numbers into L' . The second column of D consist of original frame numbers which are the source for duplicated frames. Lines 12–14 assign the type of forgery as forgery deletion if L' is empty, i.e. tamper frame localisation is not possible in case of frame deletion.

Algorithm 2 tries to identify the exact matching pair of every frame in the given list. Line 4 in Algorithm 2 computes the EMD correlation score for every frame in L against all other frames in the input video. The frame $v(ind)$ engendering minimum EMD score, $corr_{min}$, with the frame $v(i)$ is identified in line 7. This identified frame has a high correlation and hence produces a low EMD score with the frame $v(i)$. If the EMD score between two frames is less than the threshold, T_{corr} , (Line 8) and both the frames are at least ten frames apart (Line 3), then both the frames are reported as matching pairs. We consider ten frame differences between the frame pair because a very small frame difference produces a high correlation due to adjacency instead of the forged operation. Moreover, very small length forgery (less than ten frames) does not occur in reality, since the frame rate of real-time videos is ~25 frames per second. Hence forger needs to copy–paste/insert/delete more than ten frames to create a natural-looking forged incident (see Fig. 3).

To summarise the EMD plot analysis, insertion and duplication forgery produces start and end spikes. On checking the matching frame pair, we can distinguish between insertion and duplication forgery. On the other hand, deletion forgery produces a single spike in the EMD plot. To do a double check of the two spikes scenario, we also estimate the EMD correlation between the preceding frame of the first peak and the succeeding frame of the second peak. If this value satisfies adjacency and continuity, then we may confirm the two spikes in the plot are start and end tamper points rather than two single deletion points (see Fig. 4).

```

D = findDuplicateFramePair(v, L, Tcorr)
//Input : Video object, v
//      tampered frame list, L
//      threshold, Tcorr
//Output : List of duplicate frame pairs, D
1: for each i in L do
2:   for j = 1 : v.numberOfFrames do
3:     if |i - j| > 10 then
4:       corr(j) = emd(v(i), v(j))
5:     end if
6:   end for
7:   [corrmin, ind] = min(corr)
8:   if corrmin < Tcorr then
9:     add(i, ind) into D
10:  end if
11: end for

```

Fig. 3 Algorithm 2

```

//Input : Video object, v
//      anomalyframe list, L
//      correlation threshold, Tcorr
//Output : List of tampered frames, L'
//      forgery type, f
[L', f] = typeIdentification(v, L, Tcorr)
1: for i = 1 : length(L) do
2:   corr = emd(v(L(i) - 1), v(L(i) + 1))
3:   if corr < Tcorr then
4:     L' = L(i) : L(i + 1)
5:     D = findDuplicateFramePair(v, L', Tcorr)
6:     if D = ∅ then
7:       f = 'frameinsertion'
8:     else
9:       f = 'frameduplication'
10:      L' = D(:, 1)
11:    end if
12:  end if
13: if L' is empty then
14:   f = 'frame deletion'
15: end if

```

Fig. 4 Algorithm 3

4.7 Video recovery

It is possible to recover the original video in case of frame insertion and frame duplication forgery. Let i and j be the start and endpoint of tampering. On deleting the frames from i th to j th location, we can recover the original video. In case of duplication forgery, the tampered frames along with its matched frame pair sequence are also predicted using Algorithm 2. The recovered video satisfies the following three properties w.r.t EMD vector sequence.

Continuity: Since the tampered sequence is successive, removing tampered EMD values from the EMD vector results in a

```

//Input : Video object, v
//      tamper frame list, L'
//      peak threshold, Tpeak
//Output : Recovered video, V'
V' = recoverTamperVideo(v, L', Tpeak)
1: nof = v.numberOfFrames
2: for i = 1 : nof do
3:   if (v(i) ∉ L') then
4:     add v(i) to V'
5:   end if
6: end for
7: L = detectAnomaly(V', Tpeak)
8: if L ≠ ∅ then
9:   error('Recovery error')
10: end if

```

Fig. 5 Algorithm 4

continuous vector with no abrupt changes. In reality, continuous multi-frame videos are more informative than discontinuous short-length framed videos. Hence continuous videos are more likely to be forged than sparse videos. Due to this, continuity is an ensured property in original videos. Therefore, recovered video should not produce sudden spikes in the EMD plot.

Adjacency: In general, the difference between adjacent EMD values in the recovered video is nearer to zero. Adjacent frames usually have high similarity because of video consistency which is the main property of original videos.

Steady: The EMD score values for recovered video rises or lowers steadily. None of the elements in the vector are outliers.

The steps in recovering the video are described in Algorithm 4. The tampered frames are removed from the forged video to recover the original video. The recovered video is tested for anomalies in line 7 of Algorithm 4. The recovered video should be devoid of anomalies. If any anomalies exist, the Algorithm 4 reports error. Indeed, it is not possible to recover videos undergone deletion forgery. We will only report the point of deletion that happened in the video (see Fig. 5)

The different type of EMD plots due to different kind of forgery on a sample video ('bowing.y4m') taken from Derf (<https://media.xiph.org/video/derf/>) data set is shown in Fig. 6. The tampered sequence length is ten in these forged videos, i.e. ten frames are inserted (from another video of the same data set), duplicated, and deleted. The abrupt spikes in Figs. 6b–d prove that the video has tampered. Red markers in the plot demonstrate that there is a sudden change of flow from the previous to current to the next points. Similarly, green markers in Fig. 6a validate the continuity in the flow of EMD values for the original video. Even though peaks occur in Fig. 6a due to scene changes in the video, there is a gradual rise of value instead of a sudden transition.

5 Experimental results and discussions

In this section, we evaluate the performance of our proposed inter-frame forgery detection method by piloting a sequence of experiments. First, we introduce the data set and the metrics used for evaluation. Then we determine the best set of the parameter involved in the proposed work with validation tests on test video sequences. Finally, the results of the proposed work for three inter-frame forgeries and comparison analysis with six recent state-of-the-art methods are presented.

5.1 Data set

As most of the public data sets available are meant for intra-frame regional forgery, we simulated an inter-frame forged data set from

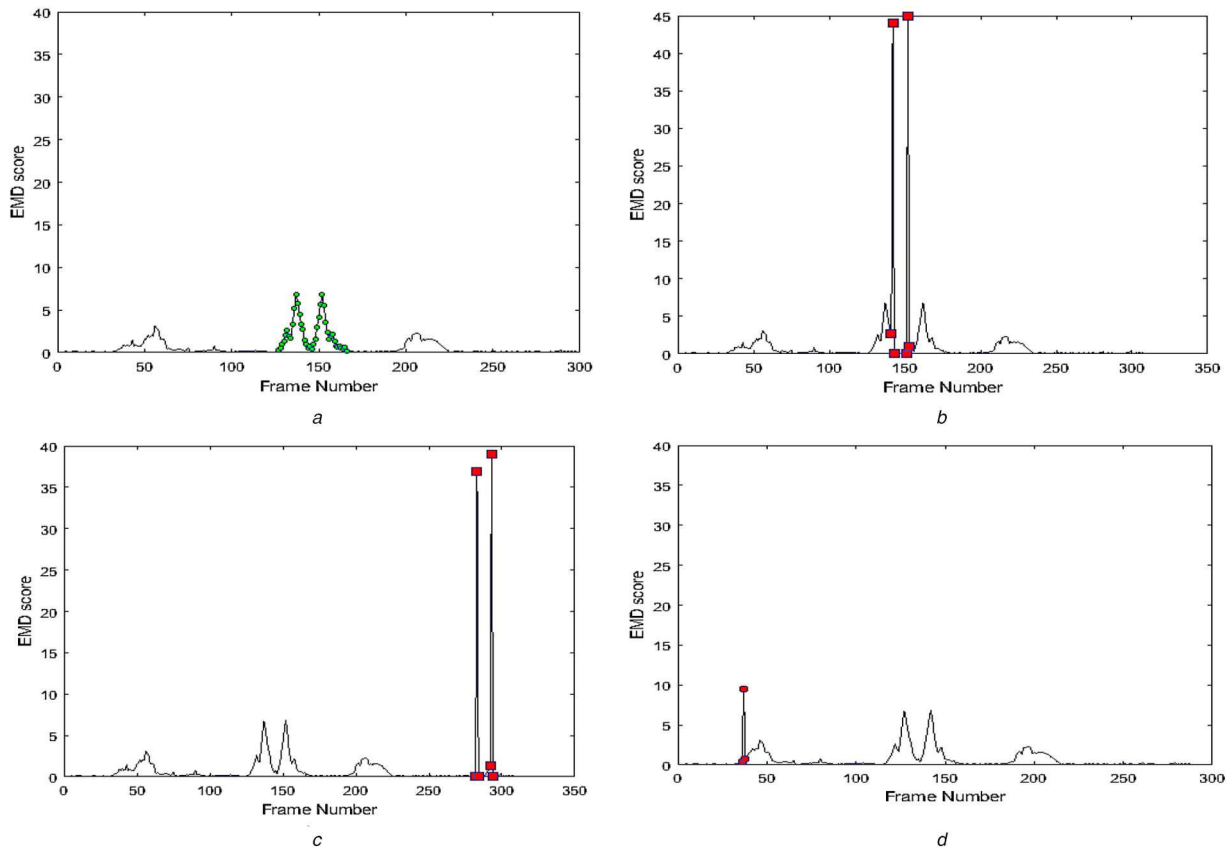


Fig. 6 EMD plot for
(a) Original video, (b) Insertion forgery, (c) Duplication forgery, (d) Deletion forgery

Table 1 Details of video data set used for evaluation

Database	Format	Resolution	Length	No. of videos	
				Original	Forged
DERF	yuv	640 × 360	90–300	20	60
DERF	y4m	1920 × 1080	60–120	20	60
YouTube	avi	varies	60–600	10	30

publicly available videos. The frame-level forgery was generated by inserting/deleting/duplicating a random sequence of frames as [12–18] did. The original videos used for tampering creation were mostly taken from a public data set DERF (<https://media.xiph.org/video/derf/>) and some of them from YouTube. The inter-frame tampering was realised in three ways: (i) inserting a set of successive frames from a source video to a random location of victim video. During this process, the source and destination videos are chosen with the same genre (for e.g. both are traffic videos) and resolution to make the tampered video more genuine. (ii) Deleting a set of successive frames from a random location of a video. The deletion is carefully designed so that it is not visible to naked eyes. (iii) Duplicating a set of consecutive frames from a random location to another non-overlapping position of the same video. To be more realistic, some tamper videos are intentionally prepared by inserting/deleting/duplicating frames in a challenging position instead of the random position. Generating more number of challenging videos is infeasible as the manual editing is complex and time-consuming compared to automatic editing.

The final data set is composed of 50 videos for each forgery ($3 \times 50 = 150$) and 50 original videos with a total of 200 videos. Among them, 30 original videos were taken from a stationary camera and 20 original videos were from moving camera, both comprising static and dynamic scenes. The frame tampering process was repeated for different numbers of frames namely 10, 20, and 40 to be inserted/deleted/duplicated. We know that one second in a video is ~ 25 frames. Therefore, the tamper length 10, 20, and 40 used for generating tamper videos correspond to 0.4, 0.8, and 1.6 seconds in a video. Hence the tampered frames

inserted/deleted/duplicated in a video cannot be discriminated visually. Tamper process was also repeated for different coding standards namely H.264, Motion JPEG and Motion JPEG 2000. The original videos are in y4m/yuv/avi format. Details of the simulated video data set are given in Table 1.

5.2 Evaluation standards

We used precision, P and recall, R as defined in (14) and (15) for evaluating and analysing the results of the proposed methodology, respectively. A high P implies a low false-positive rate, and a high R implies a low false-negative rate

$$P = \frac{TP}{TP + FP} \quad (14)$$

$$R = \frac{TP}{TP + FN} \quad (15)$$

where TP is the true positive rate (i.e. forged video detected as forged), FP is the false positive rate (i.e. original video detected as forged), and FN is the false-negative rate (i.e. forged video detected as original).

We used a 64-bit Intel i5 windows 10 processor for computation, and the programming language used for implementation was Matlab R2019a.

5.3 Parameter determination

We have used two threshold parameters, T_{peak} (see Algorithm 1) and T_{corr} (see Algorithms 2 and 3) in the proposed method. T_{peak} is used to locate sudden peaks in the EMD correlation plot, which acts as a boundary for deviation of adjacent frames. T_{corr} is used to determine the matching pair for duplication forgery. Optimal parameter value determination is an important step for achieving robustness and avoiding overfitting. Choosing a threshold value from a set of iterative values that maximises the performance may not be optimal as the results can be inadvertently biased by data set selection [29].

To experiment with the best settings for the thresholds, we have applied a four-fold cross-validation process. A subset of 100 videos (50 original + 50 forged) is chosen from the simulated data set to determine the optimum value for T_{peak} and T_{corr} . Out of the entire set of 100 videos, three-fourth of the set are randomly chosen for

Table 2 Precision and recall rates for different T_{peak} during the training phase

Parameter	Precision	Recall
1.0	0.673	0.953
1.5	0.726	0.966
2.0	0.981	0.981
2.5	0.985	0.984
3.0	0.985	0.943
3.5	0.984	0.939
4.0	0.986	0.925
4.5	0.988	0.850
5.0	0.991	0.756
5.5	0.991	0.710

Bold values indicates the best setting for the parameters.

Table 3 Precision and recall rates for different T_{corr} during the training phase

Parameter	Precision	Recall
0.05	0.971	0.995
0.1	0.982	0.997
0.15	0.971	0.999
0.2	0.956	0.984
0.25	0.863	0.922
0.3	0.818	0.868
0.35	0.755	0.790
0.4	0.693	0.763
0.45	0.688	0.724
0.5	0.650	0.687

Bold values indicates the best setting for the parameters.

Table 4 Detection results of the proposed method

Attacks		Precision	Recall
no attack		0.985	0.975
compression	H.264	0.972	0.965
	MJPEG 2000	0.981	0.971
	Motion JPEG	0.982	0.972
filtering	mean	0.983	0.974
	median	0.982	0.971
	Gaussian	0.981	0.968
noise addition	salt and pepper	0.979	0.968
	Gaussian	0.981	0.974
	Poisson	0.981	0.973
tamper sequence length	10	0.979	0.974
	20	0.986	0.974
	40	0.989	0.976

training and one-fourth of the set for testing. Four rounds of experiments were performed by exchanging the four subsets (three training and one testing) cyclically. The results from these rounds have been averaged to estimate the predictive performance. We varied T_{peak} between 1.0 and 5.5 with a step of 0.5 and T_{corr} between 0.05 and 0.5 with a step of 0.05

The precision and recall rates obtained during the training phase with respect to T_{peak} and T_{corr} are reported in Tables 2 and 3, respectively.

From Table 2, it is inferred that $T_{\text{peak}} = 2.5$ is the best setting to locate the frame which is deviating more from its adjacent frame. Also, Table 3 infers that setting $T_{\text{corr}} = 0.1$ matches duplicated frames very well in frame duplication forgery.

5.4 Robustness test

To evaluate the robustness of the proposed methodology, we have created challenging tampered videos using post-processing attacks to the videos in the simulated data set. Post-processing is the process of improving the quality of an operation finally after performing that operation. Post-processing attacks are generally applied to a video after tamper process to hide the traces of tamper operation. Common post-processing attacks are compression, filtering, and noise addition. The detection performance of the proposed work on these confronted videos is tabulated in Table 4. A random value from (10, 20, and 40) is chosen as the tamper length for these tested videos. To test our work for different tamper lengths, experiments are performed for the simulated data set by changing the tamper length as 10, 20, and 40. The detection accuracies of the proposed method for different tamper sequence lengths are also reported in Table 4. It is evident that the proposed method achieved very similar precision and recall rates as that of detection rates on videos with no post-processing attack. Due to the robustness of EMD, the performance has not been compromised for post-processing attacks. We studied the relationship between EMD flow of original/tampered videos with no post-processing attack (denoted as Type I) and the same set of post-processed videos (denoted as Type II). Fig. 7 portrays the mean EMD score for 100 Type I and Type II videos. It is evident that mean EMD scores of both Type I and Type II videos are nearer to each other which proves EMD is robust for post-processing attacks such as compression, filtering, and noise addition.

To check whether the proposed work handles scene changes in a video, careful manipulations are carried out. We took a challenging case of deleting some frames from the starting point of a scene change. In such a case, the ending point of deletion will be the starting point of the scene change and hence will be difficult to decide whether the sequence is deleted or it is still the original video. It is noted that the proposed work made the correct decision. Also, the work is tested for deleting entire frames in a scene and the detection result is found to be positive. Fig. 8 shows the EMD correlation plot for some three sample videos tampered by different

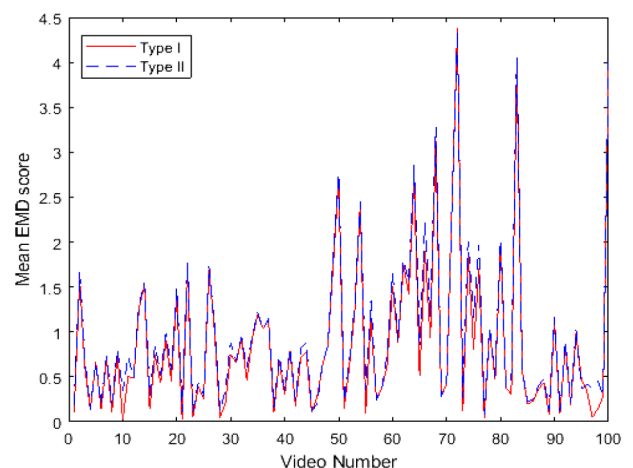


Fig. 7 Mean EMD plot for Type I and Type II videos

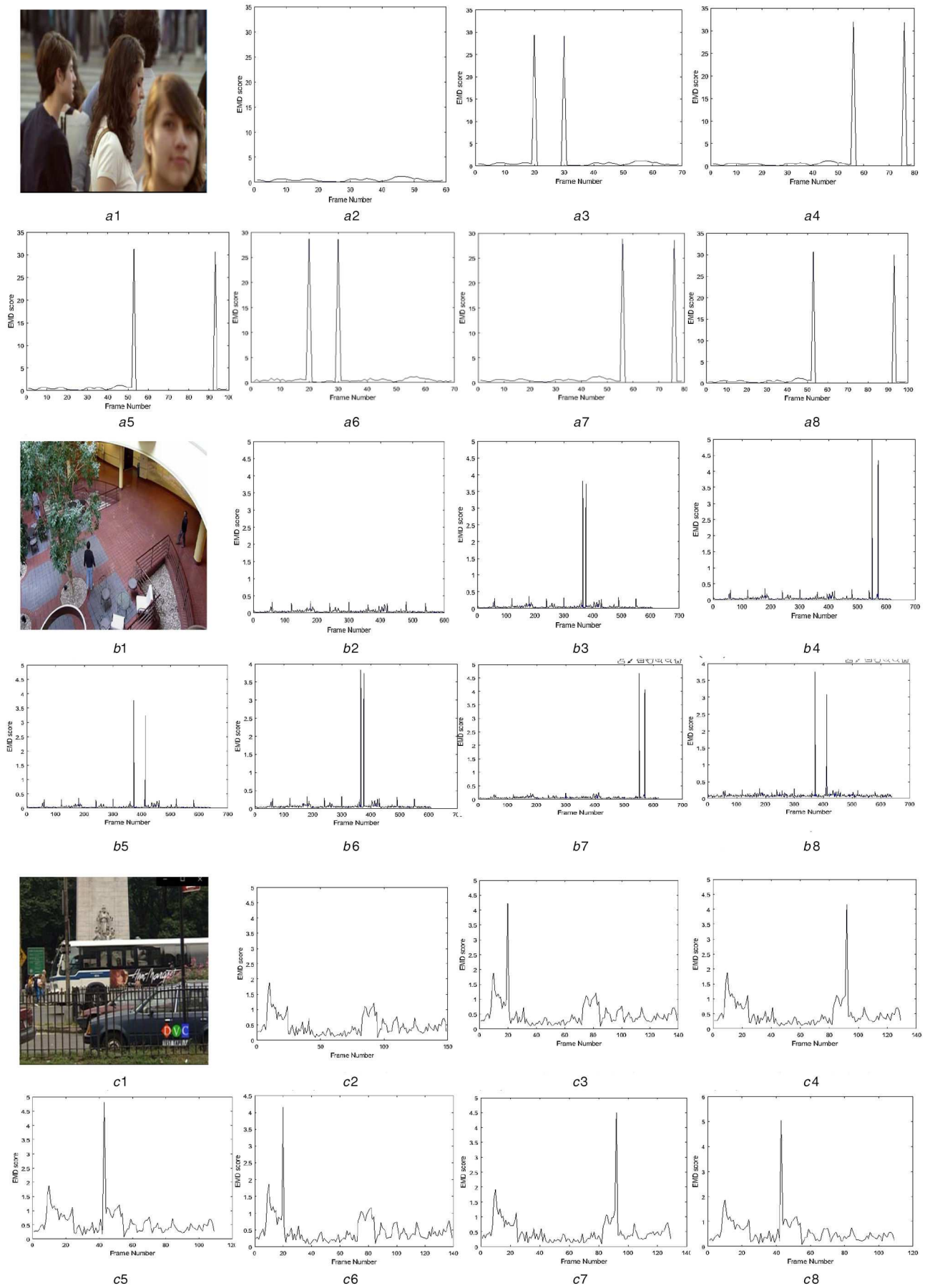


Fig. 8 Detection results for different kinds of tampering
(a1–a8) Video sample 1, (b1–b8) Video sample 2, (c1–c8) Video sample 3

kinds of forgery followed by compression, filtering, and noise addition.

Figs. 8a1, b1, and c1 contain a sample frame from the input video. Figs. 8a2, b2, and c2 show the EMD plot for non-edited

Table 5 Comparison of the proposed method with related works on a simulated data set

Work	Frame insertion		Frame duplication		Frame deletion		Execution time, s
	Precision	Recall	Precision	Recall	Precision	Recall	
Yang <i>et al.</i> [12]	a	—	0.78	0.75	a	—	350
Zhang <i>et al.</i> [15]	a	—	0.83	0.81	0.88	0.85	73
Ulutas <i>et al.</i> [13]	a	—	0.89	0.85	a	—	83
Jia <i>et al.</i> [16]	a	—	0.96	0.96	a	—	55
Li <i>et al.</i> [17]	0.95	0.94	a		0.91	0.90	93
Fadl <i>et al.</i> [18]	0.98	0.96	0.95	0.94	0.93	0.90	57
Proposed	1.00	0.98	0.98	0.98	0.96	0.96	52

^aNot in the scope.

Bold values indicates precision and recall rates obtained for proposed work.

Table 6 Comparison of the proposed method with related works on DERF data set

Work	Frame insertion		Frame duplication		Frame deletion		Execution time, s
	Precision	Recall	Precision	Recall	Precision	Recall	
Yang <i>et al.</i> [12]	a	—	0.80	0.79	a	—	315
Zhang <i>et al.</i> [15]	a	—	0.85	0.82	0.85	0.81	65
Ulutas <i>et al.</i> [13]	a	—	0.95	0.95	a	—	73
Jia <i>et al.</i> [16]	a	—	0.96	0.95	a	—	46
Li <i>et al.</i> [17]	0.99	0.97	a		0.94	0.93	85
Fadl <i>et al.</i> [18]	0.99	0.98	0.97	0.97	0.95	0.95	48
Proposed	1.00	1.00	0.98	0.98	0.97	0.96	45

^aNot in the scope.

Bold values indicates precision and recall rates obtained for proposed work.

video. Figs. 8a3–a5, b3–b5, and c3–c5 show EMD plot for insertion forgery, duplication forgery, and deletion forgery, respectively, with different tampered sequence length (10, 20, and 40). Figs. 8a6–a8, b6–b8, and c6–c8 show EMD plot for insertion forgery, duplication forgery, and deletion forgery, respectively, followed by different kinds of post-processing attacks. In the post-processing attacked videos we varied tampered sequence length between 10, 20, and 40. Fig. 8 also proves that the proposed methodology is robust to post-processing attacks.

It is observed that Fig. 8a2 shows a smooth curve as compared to Figs. 8b2, c2. This is due to the infrequent scene changes in the former compared to the latter. For a tampered video with recurrent scene variation, peaks occurring due to scene changes may confound tamper anomalies. However, the peaks due to scene changes are smaller in magnitude and progressively increase.

5.5 Performance comparison

Table 5 compares the performance of the proposed work with six-related methodologies in terms of detection accuracy and execution time. All these works are implemented and tested in our simulated data set with 50 numbers of original videos and 150 numbers (3×50) of tampered videos which is a mixture of post-processing attacked/non-attacked videos. The proposed work outperforms the existing works in terms of efficiency and robustness. Proposed work and the work presented in [18] are capable of identifying all kinds of inter-frame forgery. However, our work exhibits better robustness in terms of compression, filtering, and noise addition. Most of the YouTube videos will have been h.264 compressed when they were uploaded to YouTube. This degrades the performance of the competing methods due to compression. The proposed work and the related works are also tested on DERF data set alone after removing YouTube videos from the simulated data set and the results are published in Table 6. Although the proposed work is leading, other works [13, 16, 17, 18] are also comparatively good. So we conclude that our method is more robust to post-processing attacks than other inter-frame forgery detection methodologies and also yields good results for videos with no post-processing attack.

6 Conclusion

We propose a novel idea for the identification of inter-frame forgery using EMD metric across the frames in a video. The video is split into a sequence of frames and each frame is represented by the pixel intensity distribution in the CIE-Lab colour space and the motion flow across frames. The flow prediction is done using the EMD method and the anomaly points are identified by estimating the abrupt changes in the EMD flow. The type of the forgery such as frame duplication or frame insertion or frame deletion is identified by analysing the EMD distribution of the video frames. Also, the localisation of the tampered frames is done. The robustness of the proposed work is exhibited by subjecting the video to compression, filter, and noise addition attacks. The results obtained by the proposed work are promising and the work is compared with the state of art technologies.

In the proposed work, frame insertion, frame deletion, and frame duplication are considered as individual events. However, multiple composition of these forgeries is also possible, for example, deleting a segment and replacing it with duplicated frames. Similarly, the proposed work lacks detecting deletion of a set of frames at the start or end of the video. Inserting and duplicating a set of frames at the start/end of the video lead to a single anomaly point. Although the proposed work detects the forgery in this case due to the presence of a single anomaly point, it has no clue for forgery type identification. These challenging situations lay a path for future work.

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